



Paper id: 252370

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Subject Code: BMI202

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MCAINT

(SEM II) THEORY EXAMINATION 2024-25
DISCRETE MATHEMATICS

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt all questions in brief.

02 x 7 = 14

Q no.	Question
a.	In a group of 850 persons, 600 can speak Hindi, 340 can speak Tamil. find. How many can speak (i) Both Hindi and Tamil. (ii) Only Hindi and Only Tamil
b.	Define partial order relation.
c.	Find the generating function $a_r = 2^r + 3^r, r \geq 0$
d.	Show that the set of cube roots of unity is abelian group with respect to multiplication
e.	Show that every cyclic group is abelian group.
f.	Prepare of the table of statement in logic $(p \rightarrow q) \Leftrightarrow (\sim p \vee q)$
g.	Draw the Hasse diagram of the relation as "divides" A on the set. $A = (2, 3, 5, 30, 60, 120, 180, 360,)$

SECTION B

2. Attempt any three of the following:

07 x 3 = 21

a.	Define the function. If f and g be two functions defined by $f(x) = \sqrt{x-1}$ and $g(x) = \sqrt{4-x^2}$ then describe the function. (i) $(f-2g)x$ (ii) $(fg)x$
b.	Solve the recurrence relation. $a_r - 3a_{r-1} + 2a_{r-2} = 2^r$
c.	Prove that $f_a: G \rightarrow G'$ which is defined by $f_a(x) = a^{-1}xa$ is a group of isomorphism.
d.	Prove that the statement is tautology. $[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow [p \Rightarrow r]$
e.	Show that in a distributive lattice, if an element has a complement, then this complement is unique.

SECTION C

3. Attempt any one part of the following:

07 x 1 = 07

a.	Prove that if $f: A \rightarrow B$ is one - one onto mapping then $f^{-1}: B \rightarrow A$ will be one-one onto mapping.
b.	If R_1 and R_2 are two equivalence relations on set A, then the prove that $R_1 \cup R_2$ is also an equivalence relations on A.

4. Attempt any one part of the following:

07 x 1 = 07

a.	Prove by induction method show that $1+3+5+\dots+(2n-1) = n^2$.
b.	Solve by the generating function evaluate it, $1^2+2^2+3^2+\dots+r^2$

5. Attempt any one part of the following:

07 x 1 = 07

a.	Show that any group of elements less than or equal to four must be abelian.
b.	State and prove that Lagrange's theorem in group.

6. Attempt any one part of the following:

07 x 1 = 07

a.	Write the Boolean expression $f(x_1, x_2, x_3) = (x' x_2) (x_1 + x_3)$ in both disjunctive and conjunctive normal form.
b.	Obtain CNF and DNF of $(p \wedge q) \wedge (q \rightarrow p)$

7. Attempt any one part of the following:

07 x 1 = 07

a.	If $A = \{a, b, c\}$. Show that $(P(A), \subseteq)$ is a poset and draw its Hasse diagram.
b.	Define isomorphic lattice with a suitable example.